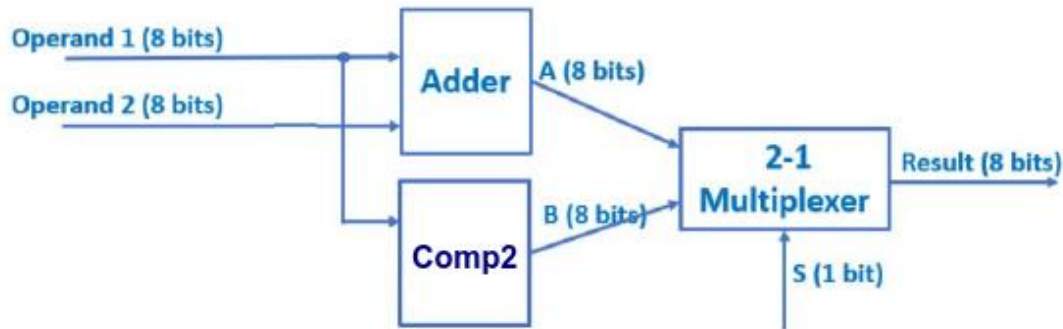


Laboratory 7 (100 Points)

Assignment 1 (50 Points)

Processors should have a separate hardware unit to perform all arithmetic operations (except the increment in the program counter). Our design contains a simple Arithmetic Logic Unit (ALU). The simple ALU has three inputs: operands 1 and 2, and the select input. Operands 1 and 2 are 8-bit inputs and the select input is an 1-bit input. This ALU also has one 8-bit output, which is the result of the computation.



1. Design a new instruction named COMP2 (two's complement). This instruction calculates the 2's complement of an operand. Draw the format of the new instruction. **(25 points)**
2. Design the ALU that can perform an 8-bit addition operation to implement the Add instruction and a 2's complement operation to implement the COMP2 instruction. **(25 points)**

Assignment 2 (25 Points)

Design a sequence detector for the bit pattern “101” which has one serial input **X**, and one output **Y**. The detector asserts **Y = 1** when the previous input value “101” is recognized; overlapping occurrences are allowed. Assume that the rising-edge-triggered clock is used.

- a. Derive the state diagram. **(5 marks)**
- b. Derive the state table. **(5 marks)**
- c. Derive simplified expressions. **(12 marks)**
- d. Design the logic diagram. **(6 marks)**
- e. Simulate the circuit with the following requirements. **(2marks)**
 - Initial state: 0 for all flip-flops
 - Input X: 101101

Assignment 3 (25 Points)

Design a **4-modulo synchronous up-counter** using two D flip-flops, one control input X, and one asynchronous reset input R. The counter counts from 0 to 3 and resets to 0, controlled as follows:

- When $X = 0$, the counter holds its current state.
- When $X = 1$, the counter increments on each clock pulse 00, 01, 10, 11.
- a. Derive the state diagram. **(5 marks)**
- b. Derive the state table. **(5 marks)**
- c. Derive simplified Boolean expressions for D flip-flop inputs. **(10 marks)**
- d. Design the logic diagram of the circuit. **(10 marks)**
- e. Simulate the circuit with the following requirements **(5 marks)**
 - Initial state: 00
 - Input X: 11111